MODULE M8E

M8G-LITE 24-LAYER ICOSAHEDRAL PRE-TEST

Opera Numerorum -- Battle Plan v1.6

David Fox -- June 2026

PHYSICS CORRECTION FROM INITIAL DRAFT

CORRECTED: k_c(H3) ~ 2.13, NOT 3.183. CORRECTED: 720 vias (24 x 30), NOT 30 vias total. k_c = 3.183 is the H4 (120-cell) Bost-Connes fixed point. The H3 (icosahedral) fixed point is ~2.13. The cliff will appear at a different voltage in M8E than in M8G.

PHYSICS: 120-CELL / 600-CELL DUALITY

The 120-cell (H4) has 120 dodecahedral cells and is the basis for M8G (120 layers). Its dual is the 600-cell (600 tetrahedral cells). Projecting the 600-cell to 3D under icosahedral symmetry (H3 Coxeter group) produces the icosahedron: 12 vertices, 30 edges, 20 triangular faces.

Via structure: 30 icosahedral edge midpoints per layer, rotated 15 deg/layer (360/24), stacked 24 layers. **Total: 24 x 30 = 720 vias -- same count as M8G, different geometry.** Layer pitch: 0.1mm (Rogers 4350B). Total board: 2.4mm thick.

k_c(H3) ~ 2.13 from the Bost-Connes KMS analysis of H3: exponents {1,5,9}, Coxeter number h=10, order |W(H3)|=120. Compare H4: exponents {1,11,19,29}, h=30, |W(H4)|=14400, k_c=3.183. |W(H4)| / |W(H3)| = 120 = number of M8G layers.

M8E vs M8G COMPARISON

Property	M8E (this module)	M8G (full build)
Geometry	H3 icosahedral	H4 120-cell
Polytope	Icosahedron (30 edges)	120-cell (720 faces)
Layers	24	120
Vias	720 (24 x 30)	720 (120 x 6)
Layer pitch	0.1mm (Rogers 4350B)	TBD mm
Board thick	2.4mm	TBD mm
k_c target	2.13 +/- 0.10	3.183
f_res	299.314 MHz	299.314 MHz
C_ratio	PENDING M22-H3	5.724
Substrate	Rogers 4350B	TBD
Fab cost	~$400	~$3,000
Fab time	2 weeks	4 weeks
Via tolerance	20 um (relaxed)	5 um

H3 COXETER DATA

H3 (icosahedral symmetry): Exponents {1, 5, 9}. Coxeter number h=10. Rank n=3. Order |W(H3)|=120. Positive roots: h*n/2 = 15. Diagram: o---o===o (the 5-bond gives 5-fold symmetry).

H3 angular signature at via centres (3D, certified by 24cell_vertices.py): 60 deg x2 per via (equilateral triangle pairs), 108 deg x2 per via (pentagon interior -- H3 signature), 144 deg x2 per via (golden gnomic pairs). Total: 30 x C(4,2) = 180 angle checks. MAX_DEVIATION: 0.000000 deg. H3_CERT: VALID.

24-LAYER STRUCTURE (Rogers 4350B)

Layer	Rotation (deg)	z_mm	Vias on layer
1	0.0	0.0	30 (template, unrotated)
2	15.0	0.1	30 (rotated 15 deg)
3	30.0	0.2	30 (rotated 30 deg)
...	...	...	...
12	165.0	1.1	30 (rotated 165 deg)
13	180.0	1.2	30 (anti-template)
...	...	...	...
24	345.0	2.3	30 (rotated 345 deg)

Total thickness: 24 x 0.1mm = 2.4mm. Via radius: 5mm from board centre. Board diameter: ~15mm (5mm via radius + 10mm margin). Rogers 4350B: Dk=3.48, Df=0.0037 at 10 GHz. Low-loss substrate chosen for high-Q resonance (Q > 10,000 required).

8-GATE STRUCTURE (7/8 REQUIRED)

Gate	Check	Target	Required
C01	CSV geometry valid	720 vias, 24L, 30/L	YES
C02	H3 symmetry per layer	60/108/144 deg	YES
C03	Layer count = 24	24	YES
C04	Via count = 720	24 x 30	YES
C05	f_res = 299.314 MHz	M1 certified	YES
C06	k_c(H3) = 2.13 +/- 0.10	H3 BC cliff	YES
C07	Cliff at correct voltage	2.13 V +/- 0.15 V	YES
C08	Via drill tol <= 20 um	relaxed (M8G=5um)	OPTIONAL

C07 special case: if C07 fails with C01-C06 passing, the cliff is at the wrong voltage. This means k_c(H3) prediction is wrong -- recheck M8C gear ratio (gear=3/6). Do NOT build M8G until M8C is corrected.

DECISION TREE

```
m8e_sim_check.py
|
+-- C01-C07 all PASS?
|
+-- YES -> Fab 24-layer PCB (~$400, 2 weeks, Rogers 4350B)
| |
| +-> Lab test (M8F-Lite): k_c, C_ratio, transit time
| |
| +-> PASS -> M8G 120-layer: 80%+ confidence. Go.
| +-> FAIL -> M8B falsified. Stop. Saved $2,600.
|
+-- NO
|
+-- C07 fails, C01-C06 pass?
| -> Cliff at wrong voltage.
| Recheck M8C gear ratio.
```

```
| Do NOT fab M8G.
|
+-- Other failure?
-> Debug geometry/fab. Do not fab anything.
```

ASSERTION SUMMARY (8 checks)

#	Assertion	Value	Status
1	f_res = 299.314 MHz (M1)	299.314159 MHz	PASS
2	k_c(H3) ~ 2.13 != k_c(H4) = 3.183	confirmed	PASS
3	720 vias = 24 x 30	720	PASS
4	Rotation = 360/24 = 15 deg/layer	15.0 deg	PASS
5	Board thick = 24 x 0.1mm = 2.4mm	2.4mm	PASS
6	H3_CERT from 24cell_vertices.py	VALID	PASS
7	|W(H4)|/|W(H3)| = 14400/120 = 120	=M8G layers	PASS
8	Budget $400 < M8G $3000	$400	PASS

CHAIN OF CUSTODY (SHA-256)

```
source SHA-256: 4ea02912e57cfda964b539e7822371f51f5dc8b1dc09ea4717aa10311ff7534f
stdout SHA-256: 1ff5e265703f76ddb294e8e178cfaabd10f36ff72ebd02b1dd5a5203134885d6
source: certificates/m8e_icosa_lite.py
stdout: m8e.out
depends_on: [M1, M22, M8B, M8C]
gate: M8E_CERT -- pre-condition for M8G fab decision
```